

SECTION B

Answer all questions in the spaces provided.

7. Sian was given the following instructions for an experiment to find the enthalpy change for the reaction below.

**Instructions**

- Weigh the magnesium.
- Place 25.0 cm³ of 0.5 mol dm⁻³ hydrochloric acid in an insulating plastic container.
- Measure the initial temperature of the acid in the container.
- Add the magnesium to the acid and stir.
- Measure the highest temperature reached.

- (a) Draw a labelled diagram showing how the apparatus is set up to carry out this experiment. You should show how the heat losses could be minimised. [3]



(b) Sian recorded the following data.

Mass of magnesium strip = 0.1 g

Initial temperature of acid = 21.0 °C

Highest temperature reached = 35.5 °C

Use these data to calculate the enthalpy change for the reaction, ΔH .



Assume that 4.18 J is needed to raise the temperature of 1.0 cm³ of all aqueous solutions by 1.0 °C and that the hydrochloric acid was in excess.

Give your answer in kJ mol⁻¹.

[3]

$\Delta H = \dots\dots\dots$ kJ mol⁻¹

(c) Efan repeated the experiment using 0.20 g of magnesium and 50.0 cm³ of 0.1 mol dm⁻³ hydrochloric acid. Although he used his data correctly to calculate the enthalpy change of reaction for 1 mol of magnesium, his answer was numerically much less than that obtained by Sian. By using the data above and that in part (b), explain why Efan obtained a different value for ΔH .

[2]

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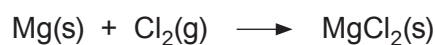
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- (d) Sian used a one-decimal place balance to weigh the magnesium. Calculate the maximum percentage error in the mass of magnesium she used. Show clearly how you obtained your answer. [2]

Maximum percentage error = %

- (e) Burning magnesium, when placed in a gas-jar containing chlorine, forms magnesium chloride and energy is released.



Suggest why this reaction cannot be used to measure the enthalpy change for the reaction that forms magnesium chloride from magnesium. [1]

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